

# Amit Kumar Mohanty

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Target Role: AI/ML Engineer | Siemens Energy

## PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of experience designing, building, and productionizing Generative AI solutions - including RAG pipelines, multi-agent systems, and LLM integrations - as scalable FastAPI microservices. Strong Python engineer (NumPy, Pandas, scikit-learn) with hands-on MLOps on Docker, Kubernetes, and CI/CD across AWS and GCP. Experienced in prompt engineering, retrieval optimization, LLM evaluation (accuracy, latency, cost, safety), Document AI (OCR/extraction), and Responsible AI guardrails. Skilled at rapid prototyping to production in cross-functional Agile teams, translating problem statements into reliable, enterprise-grade systems.

## EDUCATION

**K.R. Mangalam University, Gurgaon**

July 2021 - July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML)

**Birla Institute of Technology and Science, Pilani (BITS Pilani)**

July 2026 - July 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

## EXPERIENCE

**Research Associate - AI, Keywords Studios India - Gurgaon, Haryana**

May 2025 - Present

- Built multi-agent GenAI applications with agent orchestration, tool use, and planning/execution workflows, improving task-completion reliability.
- Engineered production data pipelines (ingestion, transformation, quality validation, versioning) delivering 85% of self-generated training datasets; awarded "Rookie of the Year 2025."
- Defined evaluation metrics and structured test reviews with a global team to harden agent policies and improve action accuracy.

**Artificial Intelligence Intern, Infosys - Remote**

Nov 2024 - Feb 2025

- Built LLM integrations for an AI knowledge/assistance system with LangChain, raising response accuracy by 35% and optimizing prompt strategies.
- Defined LLM evaluation and guardrails (automated scoring/tracing) that reduced hallucinations by 20% and improved safety of outputs.
- Optimized deployment for latency (-25%) in cloud services, collaborating across a 12-engineer team using Agile/Scrum.

## PROJECTS

**Dynamic Screen Companion - RAG + Document AI**

June 2025 - Present

- Designed an end-to-end GenAI system: LLM integration, RAG retrieval, and a FastAPI microservice backend handling concurrent inference streams with event-driven flows.
- Built the embeddings + vector-search layer (indexing, chunking, retrieval) and Document AI/OCR pipeline maintaining 95%+ extraction accuracy; containerized with Docker + CI/CD.

**Diabetic Optiscan - Vision Transformer (Peer-reviewed research)**

Jan 2025 - June 2025

- Built a Vision Transformer (ViT-B/16) on the APTOS dataset for image classification/anomaly detection; co-authored a peer-reviewed paper.
- Created a custom wavelet technique improving detection precision by 50% over legacy methods.

## SKILLS

**Programming & APIs:** Python (typing, packaging, testing), NumPy, Pandas, scikit-learn, FastAPI, Flask, SQL, Git, GitHub/GitLab

**GenAI, RAG & Agentic AI:** LLMs, RAG (indexing, chunking, reranking), Prompt Engineering, LLM Integration, Multi-Agent Systems, Agent Orchestration, Tool Use, Guardrails, Model Selection & Fine-tuning (LoRA/QLoRA), Document AI (OCR, classification, extraction)

**Frameworks:** LangChain, LangGraph, Semantic Kernel, LlamaIndex, Hugging Face, Transformers, PyTorch, TensorFlow  
**Vector, Data & Cloud:** Embeddings, Vector Databases (FAISS, Pinecone, Chroma, Azure AI Search), Data Pipelines, MongoDB, Azure OpenAI, AWS, GCP (Vertex AI)

**MLOps & Architecture:** Docker, Kubernetes, CI/CD, MLflow, Deployment & Monitoring, Microservices, Event-Driven Design, API Security, Scalability, Concurrency, Latency/Cost Optimization

**Responsible AI & Collaboration:** Responsible AI Guardrails, Bias Mitigation, Data Privacy & PII Handling, Access Controls, Auditability, Agile/Scrum, Jira/Azure DevOps

## CERTIFICATIONS

- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.
- ISRO - AI/ML for Geodata Analysis (applied ML/DL to remote-sensing imagery; time-series & pattern analysis).
- Actively pursuing Azure AI and cloud MLOps certifications.